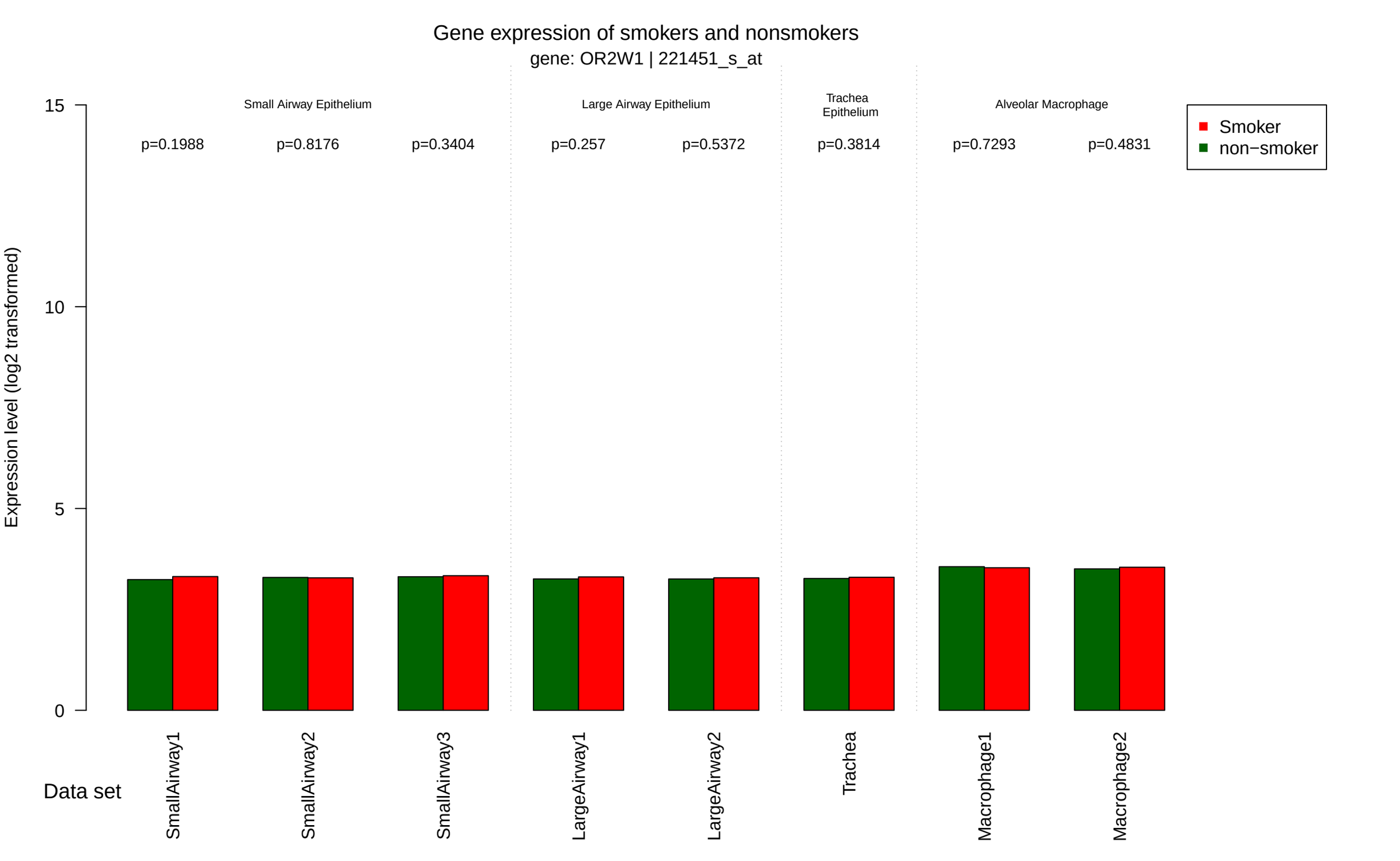
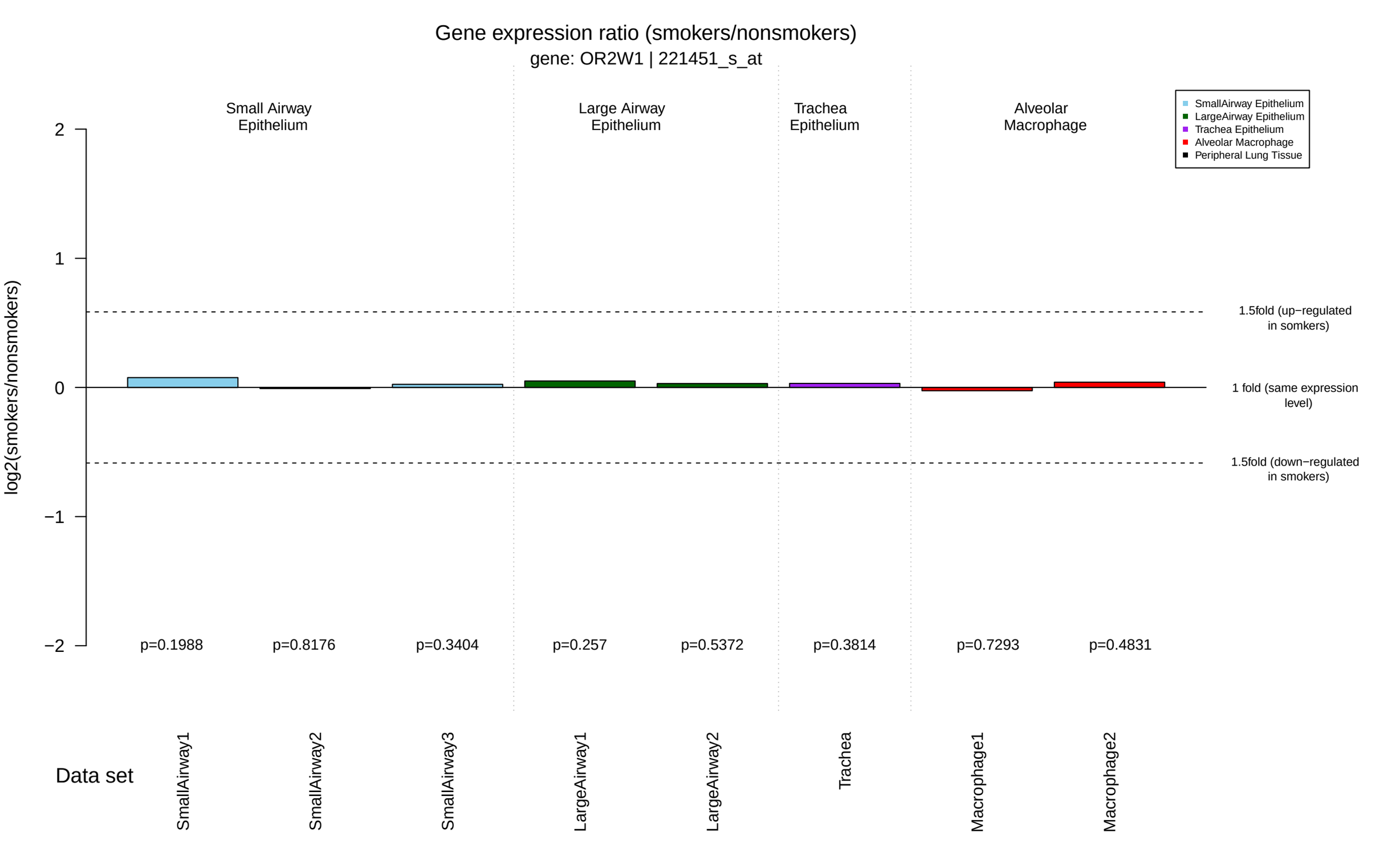
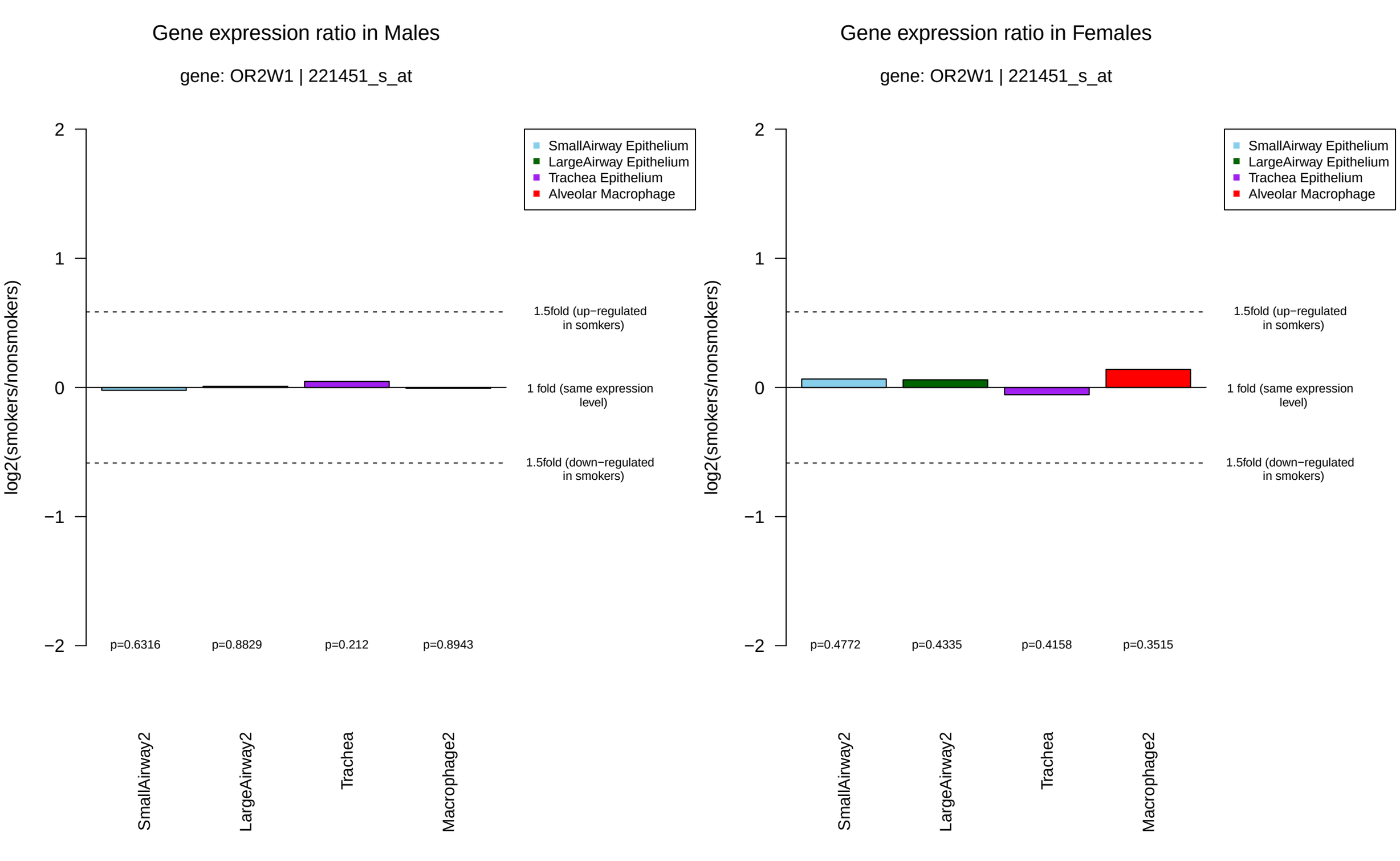
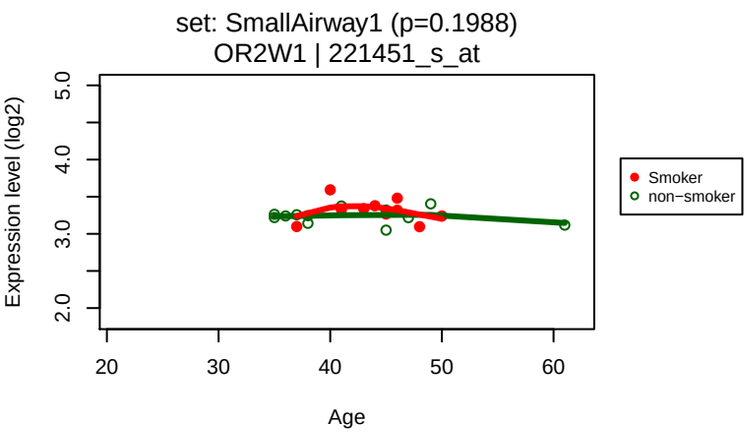
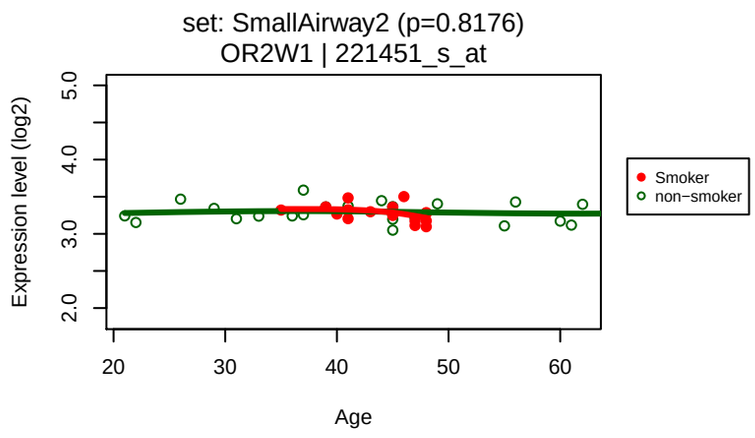
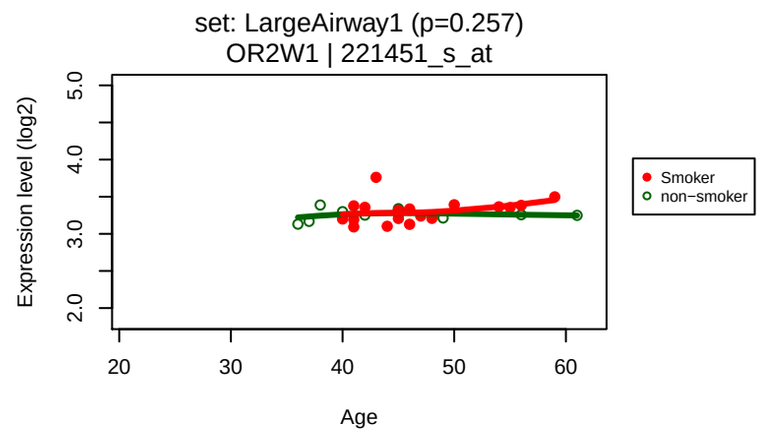

Gene expression of smokers and nonsmokers
gene: OR2W1 | 221451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1988
p=0.8176
p=0.3404
p=0.257
p=0.5372
p=0.3814
p=0.7293
p=0.4831
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

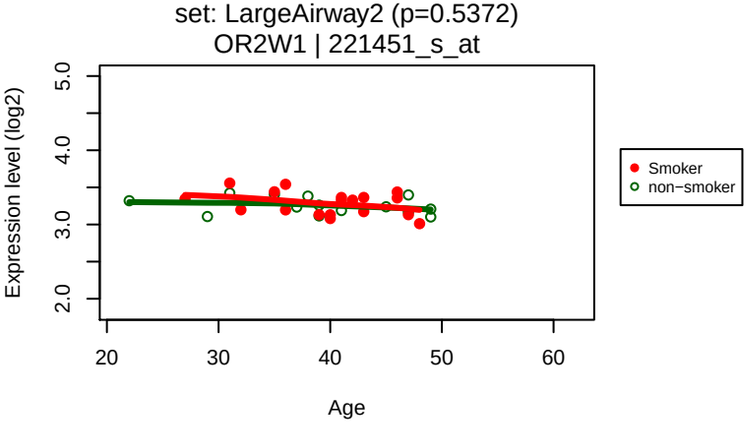

Gene expression ratio (smokers/nonsmokers)
gene: OR2W1 | 221451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1988
p=0.8176
p=0.3404
p=0.257
p=0.5372
p=0.3814
p=0.7293
p=0.4831
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

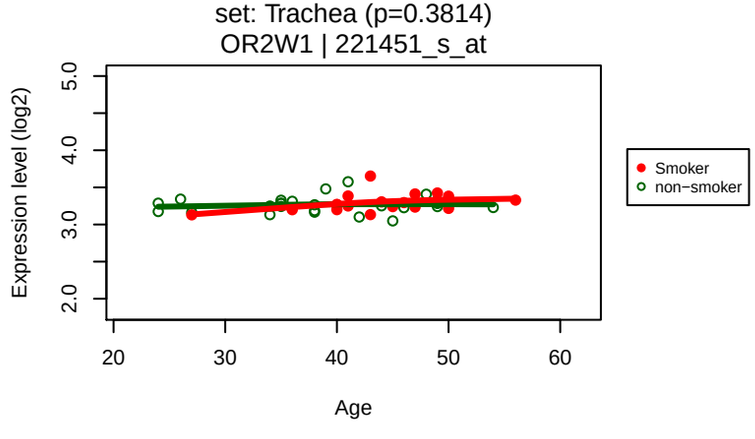

Gene expression ratio in Males
gene: OR2W1 | 221451_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6316
p=0.8829
p=0.212
p=0.8943
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR2W1 | 221451_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4772
p=0.4335
p=0.4158
p=0.3515
SmallAirway2
LargeAirway2
Trachea
Macrophage2

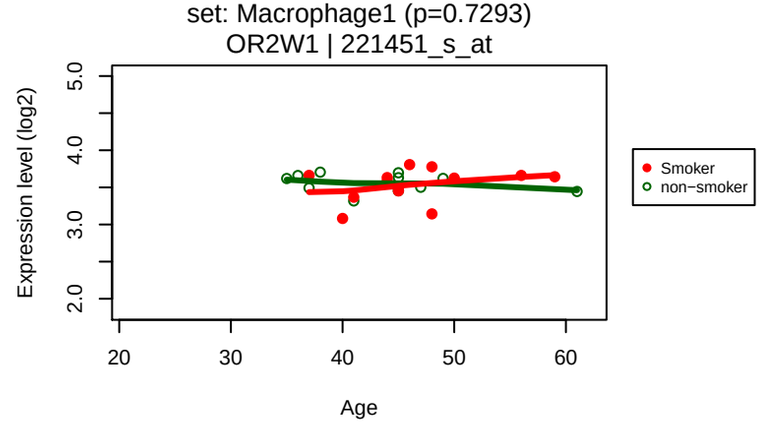

set: SmallAirway1 (p=0.1988)
OR2W1 | 221451_s_at
Expression level (log2)
Age
Smoker
non−smoker

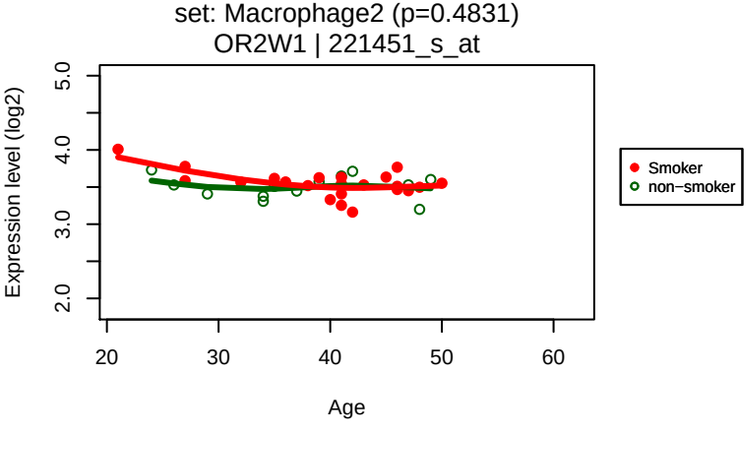

set: SmallAirway2 (p=0.8176)
OR2W1 | 221451_s_at
Expression level (log2)
Age
Smoker
non−smoker

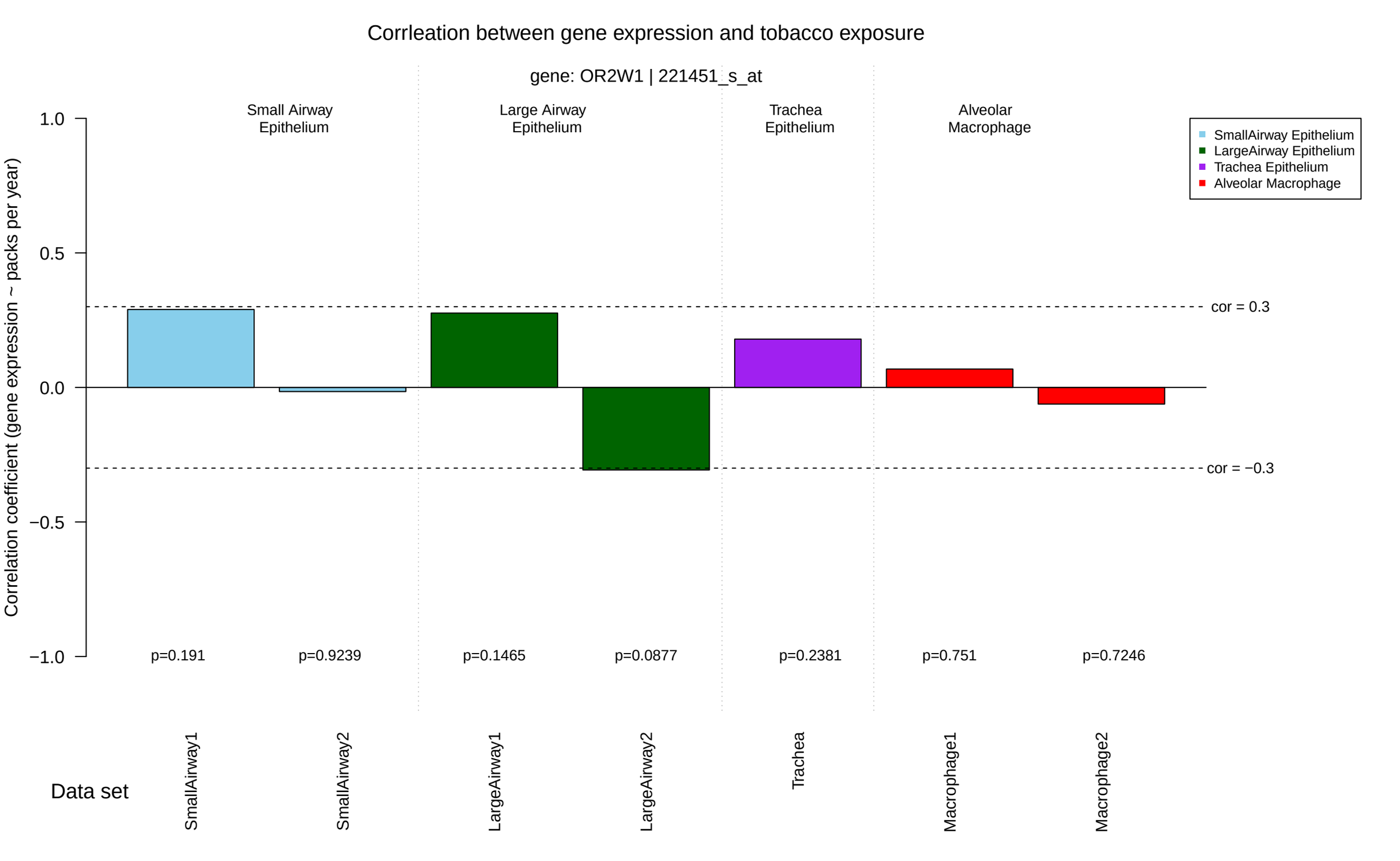

set: LargeAirway1 (p=0.257)
OR2W1 | 221451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5372)
OR2W1 | 221451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3814)
OR2W1 | 221451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7293)
OR2W1 | 221451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4831)
OR2W1 | 221451_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2W1 | 221451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.191
p=0.9239
p=0.1465
p=0.0877
p=0.2381
p=0.751
p=0.7246
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2